

ActInf GuestStream 104.1 ~ Tadahiro Taniguchi: Collective Predictive Coding

GuestStream | Tadahiro Taniguchi: Collective Predictive Coding, Active Inference | 1:14:10

Hello, welcome everyone. It is the 22nd of April 2025. We're in active inference gueststream

104.1 with Tadahiro Taniguchi. We will be hearing a presentation and a discussion on collective predictive coding and active inference. a new perspective on emergent communication. So, thank you for joining and looking forward to the presentation. Okay, thank you very much.

I thank you very much for giving me this great opportunity and speak at this great live stream. So, I'm Taniguchi from Kyoto University, Japan. Okay, today I will talking about collective practical coding and active inference a new perspective on emergent communication. So maybe collective maybe uh for many of you the corrective predictive coding maybe a new term and I would like to uh introduce the this our theory. Okay. So before starting let me introduce myself.

I'm and actually the uh I have been working on the simple emergence for a long time and but in more pro sense the AI and robotics and especially cognitive developmental robotics is my background. Okay. Last year I moved to Kyoto University and I'm now the professor at uh Koto University in Japan. So if you have a chance to visit is a beautiful city, please stop by our laboratory. Okay. Anyway, uh let me start. So before uh starting uh let me

let me share the key key points of today's talk.

So I will propose the collective predictive coding CPC in short as a dynamics underlying symbol emerging system. I will explain what is symbol emerging systems later as well and present a formulation of uh ranguage emergence based on probabilistic generative models showing its relationship between to the free energy principle and active influence.

So also position CBC as a natural generalization of FP to the group or societal

level and presents the CPC hypothesis as an interpretation of why large models appear to understand the structure of the physical world and also introduce new perspectives uh derived from or inspired uh by CBC.

BC such as CBC as a model of science and quad process theory. Maybe the topic will maybe be too much but uh I would like to introduce our

idea. I hope it is exciting. So and before diving into today's main topic maybe the CPC as I mentioned CPC will be a uh new term for you and let me introduce the corrective predict coding CPC hypothesis briefly but this uh hypothesis extends the idea of predictive coding to social and linguistic phenomena.

So it states that simple systems including language are formed through collective the predictive coding by humans. So in other words simple

emergence or language emergence can be understood as a social representation learning process. So as a result the structure of the world may be encoded in distributional semantics uh in language. So the hypothesis also suggests that this is why appear to understand the world they inherit this structure through vast amounts of distributed human genetic text. Okay. So and the main topic is CPC and the free energy principle and so on. But uh let's let me choose the LM as a starting point of today's talk maybe because everybody knows recently. Okay. So um we have but what are they actually based on? So they are completely reliant on human language data. Okay. So let's take a moment to reflect on our linguistic lives our language human language. So actually as I showed on this slide on this slide. So each individual we uh possesses a unique set of uh linguistic and the world knowledge but no one has complete knowledge or master complete knowledge of language or knowledge of the world. So and actually we dynamically learn even when we when we were children or infant. So we directly learn and adapt our linguistic knowledge through communication and including reading, writing and conversation. So rockage system itself is not a concrete uh fix system. So that is always changing and that is at some have say above

ourselves. So nobody has complete the language system. Importantly uh through uh communication the language system as I said the itself evolve over time.

uh in recent human history the vast amount of linguistic data have been accumulated in massive data sets uh on the internet or server. So by training uh by training on this corpora using next prediction and uh also of course we can use some reinfor learning method.

The large neural networks have internalized the distributional structure of human language. As a result exhibit exhibit surprisingly high level of uh language understanding and usage. uh this achievement makes a major discovery of the past decade and indeed a significant milestone in a human history I think but but as I mentioned so this capability is very much relying on our language that we created. So the question remains how does humanity form language in the first place? How do we share meaning without directly accessing the each other's minds? How did society create a language system without any centralized learning process? These profound questions motivated us to engage in constructive research on language evolution and language learning and symbol emergence. And actually the I have been we have been working on the this kind of constructive study uh since around 2010 the our community has been studying simply emergence in robotics se in

short takes a constructive approach to cognitive system inspired by developmental robotics. So actually we developed many unsupervised learning methods and artificial cognitive systems for multimodal object categorizations and concept formation language acquisition from speech signals and so on. So better to better understand human developmental cognitive systems and the emergence of language. So actually recently this kind of self-supervised unsupervised learning uh or margin model data is very very common recently but uh 15 years ago it was very very characteristic research uh to my understanding to my knowledge. Okay. So and also the most of our models were based on probabilistic generative models and predictive models. In other words, we uh okay and we defined the symbol emergence in robotics in this research series of research as constructive approach to symbol emerging systems. Okay. And as a part of this animal research we have been working on uh not only you mean the the working on the world models and the predictive coding as well in collaboration with co-authors from the active inference community. We have published a several papers on this topic. So for example we call that world models and predictive coding for cognitive and developmental robotics along with alongside uh robotic neuroscientists and cognitive scientists and we also contributed to uh a paper on world model learning learning and inference where professor carson served

as a first author. Okay, these are the kind of uh background of my research before getting into the CPC like topic. All right. Uh let me briefly uh let me begin let me briefly introduce the simple emergence system the term another term important term in this talk. So this this figure provides an overview of the concept but let me um briefly introduce the main idea here. The system does not means artificial system or a robot or something like that like that. The system here refers a complex system that enables symbol systems uh uh including language to emerge. The core idea is that the symbol emergence involves a two key dynamics happening simultaneously. So you have individual agents learning internal representations forming concept through physical interaction with the uh with the world. Uh it it kind of embodied the interaction the active interaction with the world. The traditional studies in development around cognitive robotics and also our symbol emergency robotics focused on this individual representation learning process and still many AI studies are looking at the individual capability uh of the language learning and other skills. But uh at the same time these engines interact socially through semiotic communication communication using symbols or signs including language ex and it's uh through this

social communication that uh share that that agent share the symbol systems and uh through that process symbol systems emerges across process group and this emergent system then influence the emergent symbol system then influence individual perception communication and behaviors in a top down manner. It's a kind of top down constraint. So the uh SCES simple emerging system mark emphasizes the interplay between individual learning and the collective the behavior and kind of bottom up formation and top down top down uh constraints. This kind of micro group is the heart of the simple emerging systems and I believe this is the crucial part of our language.

The after proposing this discrete lifted model from the viewpoint of emergent systems, I have been studying this system to uncover the fundamental mechanism of the human cognitive and collective intelligence in terms of symbols and language. Of course, of course the there are bunch of there are so many as related prior studies.

The researchers have been studying constructive approaches to language evolution and emergent communication for a long time. So for example in 1990s Lucas in Europe and other researchers pioneered uh language game experiment you know where agents form sign through interactions. So many algorithms are proposed and most of them are based on

language game. So the two agents or several robots play some sort of communication uh including the naming game or differential games and through that process they become sharing the signs and uh you know symbols or form language or something and so and more recently and after the you know after the after the deep learning became popular and deep learning deep reinforcement learning based approach have gained gained popularity and uh that changed the landscape of the emergent research on emergent communication. Okay. But uh basically the how to say the central idea was similar I think is similar and uh and uh you know I can I can talk about the kind of pros and cons of this approach but uh uh you know the because of time constraints let let me simply introduce another line of research our approach on emergent communication.

based on genetic model, probabilistic genetic model which has been employed in a series of study on simple emergence in robotics our uh say our uh research and uh say the answeric based approach okay so I mentioned the main game so I didn't uh talk about the example but uh uh we extended or we changed the theoretical basis of naming game and proposed metropolis hastings naming game a bit complicated name but uh that has a reason the why we call this metropolis hastings naming game uh this is a kind of game language game built on a solid mathematical foundation okay let me

introduce how does it

Back first uh there are two agents.

Consider agent A as a speaker and agent

B as the listener. The both agents both

agents observe the same object here. So

this means uh they we assume joint

attention. Joint attention is the

fundamental basis of the our language

acquisition. Basically the when infants

uh infants

around five nine months

uh old become able to perform the joint

attention and the joint attention is

believed to be the basis of uh language

learning. So I think this is a very uh

very natural

assumption. Okay. The next speaker names

the object

probabilistically and as that the naming

is uh performed by sampling from the

posterior distribution of a name based

on this image and the latent variable in

that representation.

The listener then decide whether to

accept the naming based on a certain

probability uh which depends on the

belief state. Specifically, this

probability reflect how well the

listener's belief matches the name uh

proposed by the speaker. So this agent

does not find the incoming sign

believable.

So it can reject the proposed naming

that this ability to reject uh is

crucial. After communication the

listener updates the internal parameters

related to representation learning and

naming. Then uh they switch ro and agent

B becomes a speaker and the game continues. Overall this is a very very simple game but it captures essential dynamics of how shadow symbol can emerge through communication. Okay. Uh let me add let me continue the explanation from the mathematical viewpoint. Okay. If the acceptance decision is made according to this probability this probability uh we can we can prove we can mathematically prove that the name game is equivalent to the metal processing algorithm of basian inference of the latent variable shown in the figure on the right this w let's begin with a graph with this graphical model Here O_A and O_B uh represents a visual observation of each element. Of course they can be the multimodal uh information uh that can involve other sensor input but anyway uh they have observations that is internal representations a latent variable. We also assume the existence of a variable W here which serves as a prior of these two observation two agents. Okay. uh if we treat W as a latent variable this graphical model becomes becomes very similar to multimodel or multibu variational autoencocder or some multimodel representation learning system.

So and if the two agents two agents brain were directly connected and could use both observations

simultaneously. Okay, they could jointly update the shared variable W and this variable would encode the combined information from both agents input. However, in reality they are independent cognitive systems. Do we humans cannot directly connect our brain? So let's introduce a trick

uh

algorithmically algorithmically we can decompose the model into two part and the left hand side represent agent A right hand side represents agent B. Then we consider a alternative inference method for W . So now think about the sampling process. Agent A propose a candidate latent variable W_D by sampling from its posterior distribution and

uh and uh this value is passed to agent B and who then decide whether to accept the sample based on a certain probability. If acceptance probability is designed appropriately, this process becomes equivalent to metropolis hasting algorithm which uh some of you may recognize as a markov chain method under certain condition and this uh guarantee that the sign are formed and that the categories emerge uh among agents to approach approximate the posterior distribution based on agents uh observations because this is complete the same as much uh model variational autoencoder

mation autoencoder like s like say okay

uh this is the kind of the theory the

basic basic theory and of course we can test this kind of system on some data set okay the simplest one is always aminest. Now let's move on to the result of the experiment. Uh we conducted a small experiment. Agent A observes the amun data set and agent B observes rotated uh amun data. You know in in for example in learning process even even though we have joint attention when mother is pointing oh this is zero or this is one or this is two the infant or child look at the object uh from a different direction. This is imitating such kind of situation. Okay. and uh we let the agent play the meta voice h6 naming game. Okay, this is simple result. So uh simply uh this result if the agent do not perform communication the performance is like this no communication setting the and if the agent perform the metropolis async naming game they can see a sign they can share a sign and actually the performance of clustering is improved. So this is because in the learning process the categorization process invol involving representation learning process depends not only on single agents observation but also on both agents observation the information is rich and the result becomes better. So this is wrong that that don't forget that uh this this metropolis hing naming game is a inference process of this march view

uh march view uh representation learning system of course with a single view.

Okay. So without communication with the the the agent can only use the image data. the agent observes but through the emergent communication they mathematically they theoretically make use of the information as another agent of day.

Okay, this implies that the similar emergence in society is not only for emergent communication but also for better representation learning in better understanding of the world and that's a very important implication I think so in that paper uh I published two years ago we conducted the experiment on slightly different set and I a similar result was obtained observed

Actually if we are based actually if we are based on this kind of modeling this gave us a different view of simotic communication communication using symbol

signs. Here the communication is interpreted as an interpersonal kind of crossodal influence.

First agent A observes the object and infer the latent variable. The sampling and sampling samp

uh the

what and uh and after that based on a given sample the receiver can reconstruct the

observation. If the agent perform the MH metrop naming game, they shall sign and also agent B the after receiving that message successfully

reconstruct deconstruct yeah reconstruct
the image of the
fruit. Okay. And interestingly please
see the old acceptance condition. All
acceptance condition means the listener
agent never reject the incoming sign.
Always they listen to the speakers the
and her say uh learn the learn something
from their level their uh
word. Interestingly, if we make the
agent accept all samples and the agent
do not use their belief states for
rejection, they could not successfully
form nice representation internally. As
a result, they cannot bring the images
from science. It's very interesting. I
think this uh because this is because
the communication without rejected
decision diverse from the theoretical
posterior sampling the basian influence
approximate basian influence. Okay. And
this is kind of first step the of the
our
journey. And following this the study we
have continued to explore the metropolis
hastings naming game and emergent
communication based on this kind of
robust genetic model modeling. The
recent research has introduced several
key extensions to improve scalability
and structure and adaptability
including expansion to multiple agent
not only two but three or four or more.
The expansion to compositional language.
In the previous scenario the agent just
use kind of the A or B or she
categorical categorical signs. But uh
now the agents can use uh uh say the

sequence of tokens like sentence not only has discrete sequence. So actually the the agents can use continuous signals as well.

So uh also the you know uh the the uh in emergency communication not only in this kind of naming or captioning something like that. So cooperation is important. So the one of the main field of emergent communication is uh much agent reinforcement running with emergent communication. So actually the thanks to the probabilistic modeling and the controllers as inference framework you know now the remote learning model based learning can be interpreted as a proistic inference. So we successfully integrate our idea to merge agent reinforcement learning as well. So and we can use this kind of emergent communication for uh for uh making agent run cooperative behavior as well. And also uh we we our idea our original idea was based on metropolis hastings uh algorithm that is a kind of approximate basian inference and of course metropolis hastings algorithms is not the only solution for such kind of uh implementation. So uh we are now developing a variational inference based and contrastive learning based approach and they are very they can get a similar result uh with uh this and let me introduce the one very very decent work and actually the the we uh we uploaded the display print uh one or two weeks week weeks ago and the

the Okay. Now we want to the use we want to connect our emergent communication like research to the current recent trend the of

AI and uh we the how say we replace the visal autoencoder and the gaussian mixture model like uh uh the previous model with by the visual language model visual language model. So naming naming can be replaced by captioning you know the agent can observe the natural image and pray captioning you know.

So uh mas at all extended the metropolis hing naming game to image captioning using vision language model. As they introduced we introduced a language model GPD2

uh into the capture generation process. In this scenario agents were pre-trained on different data sets and attempt to adjust their expression through communication.

But more specifically, one agent is was trained with cocoa data set and another agent uh learned uh how say trained with uh conceptual captions. So very different the way of captioning you know type of expression they have. So and but uh through the process the their way of captioning

uh is become become similar became similar you know and uh I'm not I will not get into the detail uh today but uh if you are interest if you are interested uh please take a look okay and also as we found that uh this kind of process Yes, even prevent the catastrophic forgetting that was a

problematic uh how say problem in the fine tuning in BM as well this may be a uh engineering aspect but uh uh I think uh it

has much implication anyway okay these studies along with the idea of viewing symbol or language emergence as decentralized patient

have led us to new concept and that is a collective coding of

CPC. Uh the idea is something like that.

So through the creation and negotiation of symbols or symbol systems societies may optimize collective world models.

This perspective suggest a social extension of the free energy principle.

Uh later I will talk about this a bit more uh in detail.

Uh okay. And uh and uh I think this is very much inspiring

idea and uh I think so human

society is operating the same mechanism same things and uh I'm I'm

wondering actually uh human when when humans play. When people play the

similar naming game, are they following the similar acceptance ratio or

rejection ratio? I mean that if they

follow the similar the rejection ratio

acceptance ratio that shows we are

humans can perform the basia inference

posterior inference together as a group.

So the we've conducted the you know the this kind of exper experiment with

participants. So sometimes the in the study of semiotics.

So they use uh this kind of experimental setting. This is called experimental

semiotics and uh as a result uh the result supports the idea uh that the human language uh you know this humans follow the metropolis the uh follow the similar pattern to the metropolis hing naming game. Um this is a very very weak evidence I think but uh this uh relatively support the idea that uh human language evolution language follows a CBC like mechanism to a certain extent I think okay so and uh and uh let's okay so we have been building many algorithms the with the idea of that emergent communication or a simple emergence can be viewed as a kind of the genative model generative process. So and reviewing uh these works and considering future perspective perspective uh we recently wrote a prep titled uh genative emergent communication large language model as a collective world model. Uh in this paper we expressly argue that the large language model can function as a collective world model in a sense. So let me go back to the the starting point of today's talk. I talked about the light language models learning process you know we have a simple emergence system on right hand side you know so and uh we now have the collective the idea of collective predict coding this is a kind of representation learning performed by everybody and usually when we talk about the representation learning it's often it almost

always consider it is a kind of internal representation of love. But the symbol emergence the process of symbol emergence in the process of simple emergence we actually create form the external symbol systems. So like like language and uh some subknowledge and so on and uh that that process can be regarded as a kind of social representation learning or or the representation learning external representation learning performed by multi agent systems.

So in some sense language embodying collective intelligence that is the how to say then and then there with this perspective we can now say that our language system is structurally the modeling the representing the structure of the world but this is this is not a kind of objective word it's a kind of the sum of our subjective and perceptual embodied experience. thought but as a result and by internalizing this structure through predictive modeling the LM re I would say reconstruct such kind of world model internally that is a scenario and uh and many I think that this uh I think that this is uh one idea that can uh that as as a kind of the why is is so working

well. Okay. So the let me move on to the let me re illustrate the CPC hypothesis again CPC again basic idea of emergent communication and pre predictive coding or free energy principle. First let's start from predictive coding or world

model learning. uh there is a person that he uh internally has internal representation or world models. Uh we are living with action perception loop. This is very uh how say uh this is very popular way of describing the the agent interacting the world. So and uh and uh this process is uh based this is regarded as a kind kind of inference process. Okay. But uh we humans we are so many and uh and uh but and also the this interaction this process is not uh informationally uh probabilistically independent.

So yes uh we have communication and we are influencing each other. Following the prove coding hypothesis through language here we can form uh language like social representation through the learning process. I think through this process we model and adapt to the world collectively.

So uh this is the CPC but CPC process explaining the the uh simple margin but but with this view we can change uh we can change our view until now we believe that each person is an independent and autonomous agent but let's change the viewpoint first language is here language is a kind of autonomous system here.

Okay. And uh and uh language has sensor sensors. That's us. Language uses us to model the world through interaction.

From this point, we are subordinate to the language system. Actually this kind of view is often mentioned in

sociology and some other academic fields and we are kind of the kind of you know the uh people working for language. Okay. I think this kind of dual viewpoint is very interesting to extend our view of cognition uh to the social view. Then let's move on to the free energy principle. Okay. Recently we formalized the CPC from the viewpoint of free energy principle. Based on a generative model representing CPC, we can naturally derive variational free energy. Uh in interestingly final form can be divided into two parts. The first one is an ordinary free energy term. This is a sum of the variational free energy of a number of agent.

In addition to that, an interesting term pops up. We call this the collectiveization term. This term try to align agents external representations W with each other's internal representation.

With this time the problem of minimizing individual free energy becomes the minimization of free energy of a single of a single total system. Then actually this is this free energy is not for a single agent but uh actually that this language that this total system free energy is uh defined like this and they are divided into the the individual free energy and some mysterious collective regularization term. I think this view

can extend the free energy from energy principle from the individual to the collective level with a clear relationship to simple emergence and actually this is related to the classical semiotics. I think the corrective meanization term is particularly interesting from the perspective of semiotics in semiotics that he is Charles Sanders the father of semiotics and arbitrariness is a crucial concept science can represent arbitrary object depending on language culture society and context so while this is widely recognized To human knowledge, no studies in sigotics have explicitly addressed the functional role of this arbitrariness. For the from the perspective of simple emergence, we can argue that this arbitrariness provides a degree of freedom that allow us to optimize corrective variational free energy. So this shift the individual free energy minimization problem into a collective free energy minimization framework. So in other words while we rely on the plasticity of individual brain we become a collective social intelligence by exploiting the plasticity of similar systems. So these are my current speculative thoughts on human intelligence. Okay. So before wrapping up my talk so I want to talk about a bit more a bit several uh one or two topics. So and this gave us further inspiration about the temporal aspect of

human mind. So many of you may know uh the dual process theory sometime it is called the system one and two. So uh that tells us we human have fast thinking and slow thinking. So CPC gives us an opportunity to extend this idea to group level. So uh in our paper we discuss the system 0123 a quad process theory for multi time scale embodied corrective cognitive systems. The CPC hypothesis allows us the emergent symbol system as a type of supercognitive system that adapt to its embodied environment with extremity through temporal dynamics. Okay. The first uh we have a kind of physical dynamics. The even the some of you may know like passive dynamic walker and soft robotics they shows that even without brain our body itself interact with the world the so uh so flexibly if you have a kind of the sufficient uh you your body have a sufficient mechanical property. So this is a kind of super fast dynamics and this is important in robotics and often missed in the AI studies and yes the after that the in our you know cognitive process first dynamic system one and system two and actually the recently the system one is modeled by world model and predictive learning and imitation learning and so on and the system to the language level. The kind of derivative sort that uh uh is modeled by language large language model recently. But this is how these are happening internally inside of our mind

but uh as a group we have more slow
super slow

dynamics. It is the learning process is
corresponding to the kind of the
negotiating our symbol systems changing
our customs habit in our society that is
a more slower dynamics.

So and importantly system three and
system two zero are happening outside of
our mind but uh but uh the it is super
important for the our society and
ourselves we really have to this as
system

three

the hai uh Dr. Professor Hayi who is a
fifth. Okay. And following this work we
wrote another perspective paper
exploring the formalization of
scientific activity using the CBC. The
idea of CBC is a symbolic system emerged
through the organization of external
representations as human attempt to
model the world. You know this concept
closely aligns with scientific activity
even rather than language learning as
one of the fundamental goal of science
is construct external representation
systems that describe and enhance our
understanding of the world ultimately
improving our predictable capabilities.

So and uh okay one of the key insights
is that image name in symbol emergence
has an interesting similarity to
scientific discussion. So a
representative for

uh of scientific discussion is a kind of
peer review process. You know an agent
perform an experiment, write a paper

based on their observations, submits the paper and reviewers decide whether to accept or reject it considering their beliefs, evidence and the style of paper. And uh this structurality between the image naming game and scientific discussion suggests that uh this type of overall scientific activity can be regarded as presentation learning process. This analogy leads us to the idea of CPC as the model of science. I believe this work will bring meaningful implication to philosophy of science and other field as well.

Okay. So and in this paper actually in this paper we derived the free energy of CPC. I have explained before in the previous slide. So on the base of energy for CBC framework we can naturally connect it to active influence. Finally it extends the individual expected free energy by incorporating a collective term. you know this is now can be called collective epistemic value. So this formulation allows us to consider active inference motivated or guided by emergent language uh symbol W . The future work includes the exploring this concept as active influence at the group level potentially framing it as collective curiosity or something. You know in scientific community we we are conducting research not only collective curiosity driving scientific exploration or societal adaptation. Now this is just the beginning I think and but uh I think the this kind of approach actually connecting this has been a kind of long

journey but the the connecting the
simple emergence or how to say the
collective learning process forming the
some external representation not only
just a token of a sequence of token or
category something like that that the
scientific knowledge or laws and so on.
So modeling such kind of social
phenomena is a huge challenge for us and
future AI I think future AI. Okay. So uh
that's almost all of my talk. The
summary is something like this. We
propose a corrective predictive coding
and uh and uh and uh that it was the
uh that was from the simple emergence as
decentralization influence and that was
extended to the free energy principle
and active influence uh to the societal
level and uh I mentioned the the
relationship between LM and the kind of
the the I'll say the the world modeling
and uh we introduced generative emergent
communication and related works and
lastly the I uh I mentioned several
implications in including CPC as a model
of science and system 0123 uh quadratic
process model okay uh that's all for
today thank you very much so these are
the main differences is I added this to
the to today's abstract. So if you uh
interested in uh please uh take a look.
Okay. Thank you very
much. Awesome. Thank you for that. Lots
in the
presentation. Okay. I'll I'll ask a few
questions I wrote down and read some
questions from the live chat. So what
what is the relationship between

convergence amongst agents in their
symbolic
representation and for example
cooperative social behavior or novel
research? Like what else needs to come
into play beyond just symbolic
representation conformity to
get learning, novelty, pro-social
activity or is this or is the symbol
representation do that work by itself?

Okay. Uh could you copy and paste copy
it to the

chat and uh

[Music]

it does a bit difficult to remember.

Okay. Could

you Okay. What is the relationship
between convergence of symbolic
representation and
uh narrative coherent corporate social
behavior uh Nobel
research.

Okay. That's a very important point
interesting and uh important
point and actually the maybe the
uh that her say so the answer is not a
single one. So for for example
uh regarding the cooperative social
behavior let me take the cooperative
social behavior first.

So the say the in cooperative social
behavior. So we need say and if we play
a cooperative social behavior using
language you so we need some sort of
convergence to uh let them let the
counterpart understand my intention. So
actually so such kind of how say
convergence of symbolic representation

uh symbolic external symbol system uh

allow us to

collaborate but before that we need to

construct from the symbolic system

simple system like language you know

this itself is a kind of social

cooperation cooperative behavior. So for

example, Michael Thomas the who is the

very famous the I would say the linguist

and the developmental scientist the he

put the who put much emphasis on

cooperative behavior the as a kind of

origin of language learning and

interestingly from the viewpoint of CPC

kind of the cooperative behavior is

treated as a kind of very much innate

stuff innate like stuff. So

uh can can I go back to the okay let me

go back to the metrop game. So in our uh

point of view so usually usually the the

how say the pre previous approach the

convention approach to emergent

communication.

So as forming uh forming the language is

a model as a part of kind of policy

learning or something the kind of

expressive cooperation the have

maximizing reward or something but here

oops uh sorry here the we don't have any

reward so basically the what do I we

assumed

is is the uh is the joint attention.

Joint attention and also the we assumed

this rejection or acceptance that uh

determine whether the accept or reject

the based on their own

uh belief state.

So as a result the kind of shared

language uh

cooperatively cooperatively uh shared or formed. So this is a kind of the this is uh how cooperative behavior but it's not about the the policy learning you know but uh but the kind of the process of cooperative world modeling.

So and this is a very much different uh viewpoint and this is a kind the first point of the so so in other words convergence of symbolic system representation itself is a cooperative social. So this is the first answer and the second answer is uh no research. Uh maybe this is about the CPC MS uh collective predictive coding as the model of science and yes so as you know as you know or the as we understand this is a kind of optimization process re but the we uh how say I think that most of us don't believe this process reaches share the kind of convergence you know that we are always have to struggling to minimize the free energy and take actions you know so and uh as a kind of the you know the collective behavior CPC try to minimize the I said try to uh minimize the social level collectively energy and To that end, the each researcher each researcher is something like

this. This looks like controlled by the you know scientific society but uh they need to explore explorer to obtain new data to update the the uh how say the uh simple system with new information. So and this is corresponding to the novel research that

in our society I think. Okay.

Uh is that's from my side. Thank you.

Yeah I really like this image. It's like humans as the blanket or the interface between a disembodied symbolic system and a world that evades symbolic description alone. Um, okay. I'll ask another question and then I'll look for questions in the live chat and I put it in the chat. So you brought up this dual view of neural plasticity in terms of our learning and symbolic plasticity in terms of the development of symbol systems. So with access to LLMs and computational semantics, how is human language changing or how might it change?

That's very much interesting uh topic and uh the actually the that is project I want to start I want to launch actually this this is a kind of the very uh new and uh the interesting topic and I think yeah to be honest I don't have answer now but uh with the lens of CPC I can uh how Hey uh uh I can uh have I say the discussion over this topic and uh sorry uh let me move to the genative. Okay this one this one this one. So of course the will give us a huge impact and uh but that also depends on the in what way the we we in the future we learn we run the algorithm system. Actually two years ago, two years ago the GPD4 was launched and at that time everybody started using GPD and everything every linguistic phenomena on internet is affected by single. So but recently the situation is

changing and uh the LM becomes diversified. So the situation has changed. But uh anyway the our uh how say our communication our language is is uh has been affecting by the LM and the empirical evidences show some effect has already shown uh some effects. So the the kind of the word choice uh has already be changed to some extent but uh you know the people tend to think is some thing like a area or something like a kind of the very much different from human I'll say human cognitive system you know humans and but uh you know if the m the follows based on the as how say language generated by CBC. So basically the distribution is very much how say naturalistic one you know we we a human created the language and uh and maybe that we can use this figure. So the they are behind the there are so many humans people. So just the summarize the information and give you influence. So actually that is uh her say that is similar to uh what has was happening in the past like this. So language system was uh we we shared and we our way of speaking way of writing is affected by some you know some uh how say the shared symbol system. Of course the the distribution of nature is become very much different. So maybe that leads us to different uh way of language use. But um yes and I think the I would say the you know we cannot uh we I think we cannot determine oh this is the future

or something like that but uh we can discuss the how to say the future you know uh the future changes based on using the model of the CPC you know model of symbol emergence. you know the the if we can model the the whole background of L so that may enrich the our discussion so maybe next year or next next year I will have a clear answer I don't know thank you very much question all right I'll read the next question ML Dawn wrote In the collective predictive coding hypothesis paper, the generative model does not seem to have any mechanism to capture the dynamics of the world. This is surprising since the paper keeps referring to the dynamic nature of the world repeatedly. If my observation is correct, may I suggest to add at the very least x as well as dx/dt to the generative model. This can help immensely when the world is truly dynamic. So yeah, how did any of these papers deal with potential change through time of the reference or of the semiosis? Thank you very much. uh no objection, no defense and I totally agree with you and uh actually so when I when I uh proposed the metrop name game and uh actually the uh uh CBC so we didn't yes the as the first as the first step we didn't involve the temporal nature and dynamic nature And uh it was a kind of strategic uh one. And uh we wanted to focus on the this uh how I said this part you know to connecting the you know the connecting

the collective learning or symbol
emergence process at the group level and
the representation learning in the uh
individual level. And also we have been
conducting research on the world
modeling and the predictive learning
like stuff. And so the actually we
thought that I thought that we can
extend we can invoke the dynamic nature
after that and actually that we wrote uh
and we wrote the some papers about the
kind of the uh how say the the merch the
the uh collective
predictive learning based approach
coding based approach
uh based the meth uh margin learning
with emergency communication and also uh
it's the I I we haven't published that
but uh the our students is now working
on the world model based world how said
emergent communication based on world
model and so on and of course the say uh
we haven't invented that but uh yes I
really that we want to how say
reformulate this kind of process based
on dynamical systems based approach and
uh yeah I we have a discussion over that
with our collaborator and friends that
uh yes that I think so such kind if you
if you are somebody uh some great
researchers add such kind of aspect to
this this kind of CPC like idea that was
hugely that is very great and that's
actually also why uh we I I wanted to
have said join this live stream you know
so in the active influence community
there are so many great researchers who
uh I think who can add the another value

uh or another well uh say nice idea nice combination to this kind of how say the research bridging collective behavior and the uh individual learning process.

So yes so so yes thank you very much for your

comment. Thank

you.

Um, sort of in closing, what are you looking forward to this year or how might people get involved beyond like reading the papers?

Thank you very much. So uh last year so after I got the

inspiration of shitishi that I last we rushed to write several papers and uh I wanted uh I wanted to share uh this uh

how to say uh this idea and I think that

this will be a kind of a streams of uh

this will say the how the the following

streams of researches and in Japan uh we

I I have now uh have said uh made

several uh many collaborators but u but

uh if you think this is in uh how say

this is an interesting aspect but it's

insufficient please

uh how say uh how say uh I'll say uh

please add another value or another

contribution uh to the to the to the

research community and active influence

community as well. And actually the last

year and the year before last I really

wanted to join the IW

uh AI Dubai in there workshop on active

influence and uh yes my friend uh Tessia

was a keynote speaker last year actually

because of some

universities job or something I couldn't

uh attend the meeting that this year I really want to join But if I miss the I my I'll say my collaborator or my students will join the and I hope that we can have a fruitful collaboration or discussion in the pen and not only the in person but also the online as well. Uh I welcome the any any messages or any tweets are welcome to and and I'm the kind of the long-term Twitter or ex user and uh yes and uh please uh keep in touch. Thank you very much. Thank you. It was an awesome presentation. I hope people in the community read these papers and start to build and work with them. So, I hope to keep in touch and see you again. See you again. Thank you very much. Bye.